Valid prediction intervals for transferred multivariate calibrations: reusing the transfer standards with a leave-one-standard-out jackknife+

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Abstract

Background: Calibration models built on one spectrometer or harvest season are routinely deployed elsewhere after calibration transfer (e.g., piecewise direct standardisation, PDS) or slope/bias correction, using a handful of transfer standards. Analytical results also require a stated uncertainty. However, whether the prediction intervals of a transferred calibration keep their nominal coverage has not, to our knowledge, been studied systematically, and splitting the scarce standards between transfer and uncertainty calibration wastes them.

Results: We propose a transfer jackknife+ (TJ+): every standard is left out in turn, the transfer step is re-fitted on the remaining standards, and the leave-one-out residuals, normalised by a source-learned difficulty function of Hotelling T^2 and Q residuals, define distribution-free intervals. For exchangeable standards these intervals have coverage of at least $1-2\alpha$. On three public NIR data sets (corn, three instruments; pharmaceutical tablets, two instruments; mango dry matter, four seasons) in [[n_scen_main]] pre-registered scenarios with up to 500 random repetitions each, source-calibrated intervals (ASTM-type parametric, split conformal, weighted conformal) under-covered after transfer, with mean coverage as low as [[min_cov_source]] for a nominal 0.90. TJ+N reached a mean coverage of [[tjn_cov_main]] and never under-covered for $m \geq$ [[tjn_valid_m]] standards. Its intervals were finite where the

split-standards alternative was not ([[split_inf_text]]). Robustness, ablation, ridge and 1D-CNN analyses delimit when the guarantee applies.

Significance: TJ+ provides finite-sample-valid uncertainty for transferred NIR calibrations from the standards that calibration transfer already requires, at a cost of *m* refits of the transfer step.

Keywords: calibration transfer; prediction interval; conformal prediction; jackknife+; near-infrared spectroscopy; measurement uncertainty

1. Introduction

Near-infrared (NIR) spectroscopy with multivariate calibration is a mature analytical technique [1]. Its calibrations, however, are tied to the instrument, measurement conditions and sample population on which they were built. When a model developed on a master instrument is moved to a secondary instrument, or applied to a new harvest season, its predictions degrade unless the model or the spectra are corrected [2–4]. Calibration transfer addresses this. Direct and piecewise direct standardisation (DS, PDS) map secondary-instrument spectra onto the master space using transfer standards measured on both instruments [5,6]. Slope/bias correction (SBC) adjusts the predictions using standards with reference values [2]. Domain-adaptation methods such as di-PLS and standard-free approaches remain an active topic, including in this journal [7–12].

Calibration-transfer studies are almost always judged by point-prediction accuracy (RMSEP, bias). Analytical results, however, must be accompanied by a statement of uncertainty, and multivariate calibration has a long tradition of uncertainty estimation and figures of merit [13–15]. In routine practice, approximate prediction intervals are derived from the calibration error and the leverage of the new sample, as in ASTM E1655-type procedures [16]. These intervals use the *source* calibration error model. After an instrument or season change, that error model

no longer describes the deployed predictor. The transfer step adds its own error, and the standardisation is estimated from only a few standards.

Conformal prediction provides intervals with finite-sample, distribution-free coverage guarantees under exchangeability [17–20]. In chemometrics it has been used mainly for classification [21–24] and for in-distribution regression. Examples are normalised conformal regression of mid-infrared spectra [25], conformal-inspired intervals for PLS and PCR [26], and QSAR applicability domains [27]. Coverage guarantees break when calibration and deployment data are not exchangeable, which is exactly the calibration-transfer situation. Weighted conformal prediction corrects for covariate shift [28], but an instrument change alters the relation between spectrum and property, not only the spectral distribution. Transporting calibration data between paired domains has very recently been proposed for classification [29]. We found no study of interval validity for transferred spectroscopic calibrations.

A practical constraint shapes the problem. The only target-domain data with reference values are typically the m transfer standards, often 10–30 samples, and they are already needed to fit the transfer. A valid but naïve solution splits them: half fit the transfer and half calibrate a split-conformal interval. With $m < 18$ at $\alpha = 0.1$, the calibration half is too small to give finite intervals. The jackknife+ [30] avoids sample splitting through leave-one-out refits, with a coverage guarantee of $1-2\alpha$. We use this to construct intervals for *transferred* calibrations by re-fitting the transfer step, not the source model, inside the leave-one-out loop. All standards are therefore reused for both purposes.

The contributions of this work are:

1. A transfer jackknife+ (TJ+) for transferred multivariate calibrations, with a coverage statement under explicit, checkable conditions on how the standards are selected.

2. A chemometric normalisation (TJ+N), using a source-learned difficulty function of Hotelling T^2 and Q residuals of the *transferred* spectrum, so that interval width adapts to the sample.

3. A pre-registered benchmark on three public NIR data sets covering instrument and season shift. It compares classical, source-calibrated, shift-weighted, transport-based, split-standards and Gaussian intervals, and includes ablation, robustness, model-class (PLS, ridge, 1D-CNN), sensitivity, computational and error analyses.

4. An exploratory label-free variant for the case in which the transfer standards have no reference values.

All code and the complete reproduction pipeline are publicly available.

2. Theory

2.1. Problem formulation

Let $f : R^p \to R$ be a calibration model (PLS unless stated otherwise) fitted on source-domain data D_S: master-instrument spectra or earlier seasons. In the target domain, m transfer standards $Z_i = (x_i^T, x_i^S, y_i)$, $i = 1, \ldots, m$, are available. Here x_i^T is the target spectrum, x_i^S the master spectrum of the same sample (needed only for spectral standardisation) and y_i the reference value. A transfer algorithm T maps a set of standards D to a *deployed predictor*

$$\mu_D(x) = a_D + b_D f(g_D(x)), \quad (1)$$

where g_D is a spectral standardisation and (a_D, b_D) an output correction.

- PDS or DS: g is fitted from the pairs (x_i^T, x_i^S) and $(a, b) = (0, 1)$.
- SBC: g is the identity and (a, b) come from the least-squares regression of y_i on $f(x_i^T)$.
- No transfer: $\mu = f$.

For a new target sample $\left(x_{m+1}^T, y_{m+1}\right)$ we seek an interval $\widehat{C}\left(x_{m+1}^T\right)$ with

$Pr\{y_{m+1} \in \widehat{C}\left(x_{m+1}^T\right)\} \geq 1-\alpha$, as narrow as possible.

2.2. Difficulty function

A heteroscedastic, sample-specific scale is obtained from PLS diagnostics. For a (transferred) spectrum, let $T^2(x) = \sum_{a=1}^{A} t_a^2 / s_{t_a}^2$ (Hotelling T² in the A-dimensional score space) and $Q(x) = /\!/ x_c - t\,P^T /\!/^2$ (the squared spectral residual). We set

$$\sigma(x) = max\left(\varepsilon, c_0 + c_1\sqrt{T^2(x)} + c_2\sqrt{Q(x)}\right), \quad (2)$$

with $c_j \geq 0$ fitted by non-negative least squares to the absolute out-of-fold residuals of the source cross-validation, and ε equal to 5% of their median. σ is fixed **before** any target datum is used. Within TJ+N, σ is evaluated on the spectrum after the fold-specific transfer, $\sigma_D(x) = \sigma\left(g_D(x)\right)$. Spectra that are poorly reconstructed by the source model after transfer (large Q), or that extrapolate in score space (large T^2), therefore receive wider intervals.

2.3. Transfer jackknife+ (TJ+)

For each i, the transfer is re-fitted without standard i, giving $\mu_{-i} = \mu_{D¿i¿}$ and $\sigma_{-i} = \sigma_{D¿i¿}$. The leave-one-standard-out normalised residuals are

$$R_i = ¿\, y_i - \mu_{-i}\left(x_i^T\right) \vee \frac{¿}{\sigma_{-i}\left(x_i^T\right)}, i = 1, \ldots, m. \quad (3) ¿$$

The TJ+N interval for a new target spectrum x is

$$\widehat{C}(x) = ¿$$

where $\hat{q}_\alpha^{+¿\{v_i\}¿}$ is the $\lceil (1-\alpha)(m+1) \rceil$-th smallest of $v_1, \ldots, v_m$ (and $+\infty$ if this rank exceeds m), and $\hat{q}_\alpha^{-}\{v_i\}$ is the $\lfloor \alpha(m+1) \rfloor$-th smallest (and $-\infty$ if this rank is below 1). The unnormalised variant TJ+ uses $\sigma \equiv 1$. The source model f is **never** re-fitted; only the transfer step is.

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Proposition 1. *Suppose that $Z_1, \dots, Z_{m+1}$ are exchangeable. This holds, for example, when the standards and the future sample are drawn at random from the same target population. Suppose also that T treats its input standards symmetrically, and that f and σ do not depend on the target data. Then $Pr\{ y_{m+1} \in \hat{C}\left(x_{m+1}^{T}\right)\} \geq 1 - 2\alpha$.*

Proof sketch. Fix the source model and σ, and view $A : D \mapsto (\mu_D, \sigma_D)$ as a symmetric regression algorithm acting on the augmented standards. Define, for $i \neq j$, the score of point i under the fit without $\{i, j\}$, $R_{ij} = ¿\, y_i - \mu_{-ij}(x_i) \vee ¿\, \sigma_{-ij}(x_i)$. The tournament argument of Barber et al. [30, Thm. 1] uses only the exchangeability of the matrix (R_{ij}), which follows from exchangeability of the Z_i and the symmetry of A. If $y_{m+1} \notin \hat{C}$, then $R_{m+1,i} > R_{i,m+1}$ for at least $(1-\alpha)(m+1)$ indices i. By that argument this event has probability at most 2α. Normalisation by a scale fitted together with μ does not alter the argument, because the score remains a fixed function of the point and of the fit that excludes it. ▫

Three remarks clarify the scope of the guarantee.

- **No transfer step.** When there is no transfer ($\mu_{-i} = f$), TJ+ reduces to split conformal prediction that uses *all m* standards for calibration, with exact $1 - \alpha$ coverage.
- **Minimum number of standards.** Finite upper bounds require $\lceil (1-\alpha)(m+1) \rceil \leq m$, i.e. $m \geq 9$ for α = 0.1 and $m \geq 19$ for α = 0.05. Splitting the standards in half (SCP-T-split, §2.4) needs twice as many.
- **Conditions are not automatic.** Exchangeability is a property of how the standards are chosen. Representative-design selection, such as Kennard–Stone [31], is not covered by Proposition 1 and is examined empirically (§4.4). The bound is also marginal: it holds on average over the draw of standards and the test sample, not conditionally on a particular set of standards.

2.4. Comparison methods

All methods share the same deployed point predictor μ_D fitted on all m standards. They differ only in the interval.

- **ASTM-PI** (classical parametric): $\mu(x) \pm t_{1-\alpha/2,n-A-1} SEC \sqrt{1+h(x)}$, with the source standard error of calibration and the leverage h of the transferred spectrum [16].
- **SCP-S:** split conformal prediction with the pooled source residuals (out-of-fold residuals of the source training set plus source-calibration residuals), assuming no shift [18].
- **WCP-S:** weighted conformal prediction for covariate shift [28]. Likelihood ratios come from a logistic domain classifier on the PLS scores of source spectra versus transferred, unlabelled target spectra.
- **Transport:** a regression analogue of transported conformal calibration [29]. Source-calibration spectra are mapped into the target space by a master $\rightarrow$ secondary PDS fitted on the standards, passed through the deployed pipeline, and their residuals calibrate a split-conformal quantile. It applies to paired transfers only.
- **Gauss-LOO:** $\mu(x) \pm t_{1-\alpha/2,m-1} RMSE_{\mathrm{LOO}}$, using the leave-one-standard-out residuals. This mimics recomputing the RMSEP on the standards.
- **SCP-T-split(-N):** the standards are split at random into halves. The first half fits the transfer and the second half calibrates (normalised) split conformal prediction. This is valid, at the cost of halving both sets.
- **MC-dropout** (CNN only): Gaussian intervals from 100 stochastic forward passes plus the source residual variance [32], followed by SBC.

Ablations:

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- TJ-naive-N: full-fit centre with leave-one-out residual quantile, i.e. a jackknife without "+".
- TJ-insample-N: residuals from the transfer fitted on all standards, without leave-one-out.
- TCV+N: five-fold CV+.
- TJ+N[T²] and TJ+N[Q]: σ from a single diagnostic.

Label-free variant (exploratory, LF-conv). If the paired standards have no reference values, the target error decomposes as $y-\mu\left(x^T\right)=e^S-d$. Here $e^S=y-f\left(x^S\right)$ is the source error, estimated by the pooled source residuals e_j. The term $d=\mu\left(x^T\right)-f\left(x^S\right)$ is the transfer discrepancy, estimated from leave-one-out fits on the standards as $d_i=\mu_{-i}\left(x_i^T\right)-f\left(x_i^S\right)$. The interval is $\mu(x)+\left[Q_{\alpha/2},Q_{1-\alpha/2}\right)$ of $\{e_j-d_i\}_{j,i}$. It assumes independence of e^S and d and has no finite-sample guarantee.

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